

A PROOF THAT EVERY INTEGER OCCURS INFINITELY OFTEN AS A DIFFERENCE OF BINARY AND TERNARY DIGIT SUMS

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ABSTRACT. For $n \geq 0$, let $s_b(n)$ denote the sum of the digits of n in base b . We prove that, for every $z \in \mathbb{Z}$, the equation $s_3(n) - s_2(n) = z$ has infinitely many nonnegative integer solutions. The proof adapts the fair-share step in Spiegelhofer's work on binary-ternary digit-sum collisions from the zero residue class to an arbitrary residue class. The adaptation changes only unit-modulus phases in the relevant Fourier expansion, so the established nonzero-frequency estimates are uniform in the chosen residue. The result is qualitative: no effective bound for the least solution at a prescribed offset is extracted.

1. INTRODUCTION

Write $s_b(n)$ for the sum of the digits in the standard base- b expansion of n , and set $s_b(0) = 0$. The sequence $s_3(n) - s_2(n)$ is recorded as OEIS A180017, where the question whether every integer occurs infinitely often is posed [1]. Spiegelhofer proved the zero-offset case: there are infinitely many positive integers n for which $s_2(n) = s_3(n)$ [2]. We show that the same conclusion holds on every diagonal of fixed additive offset.

The proof uses the preparation and rarefaction constructions of [2, Propositions 2.2 and 2.3]. The additional observation needed here is that the Fourier argument used there to obtain a fair share of values divisible by a parameter m is residue-uniform. Selecting the residue of a fixed target and applying the prepared digit shifts yields the desired offset. A final carry-free increment removes the possible error of one.

2. STATEMENT AND NOTATION

For $b \geq 2$ and $n > 0$, $s_b(n)$ is the sum of the digits in the unique finite base- b expansion of n without leading zeroes; by convention $s_b(0) = 0$. Put

$$F(n) = s_3(n) - s_2(n), \quad g(n) = -F(n) = s_2(n) - s_3(n).$$

Theorem 2.1. *For every integer z , infinitely many nonnegative integers n satisfy $s_3(n) - s_2(n) = z$.*

Equivalently, for every $z \in \mathbb{Z}$ and every $N_0 \in \mathbb{Z}_{\geq 0}$ there is an $n > N_0$ with $F(n) = z$. Indeed, a subset of $\mathbb{Z}_{\geq 0}$ is infinite if and only if it is unbounded. We write $e(x) = \exp(2\pi i x)$. All asymptotic statements below concern $N \rightarrow \infty$ with the target fixed.

3. THE IMPORTED DIGIT-SHIFT CONSTRUCTION

Fix $\epsilon > 0$ and, for $N \geq 4$, put

$$\begin{aligned} \lambda_0 &= \log N, & \eta_0 &= \lambda_0^{3/4}, & f_0 &= (\log \lambda_0)^{1/2+\epsilon}, \\ m_0 &= \lambda_0^{1/2}/f_0, & J_0 &= f_0^2, \end{aligned}$$

and

$$\lambda = \lfloor \lambda_0 \rfloor, \quad \eta = 4\lfloor \eta_0/4 \rfloor, \quad m = \lfloor m_0 \rfloor, \quad J = \lfloor J_0 \rfloor.$$

Let $\beta = (2J+1)\eta + 1$, and take $\nu \geq 1$ minimal with $2^{\nu-1} \geq 3^\beta$.

The following is the combination of Propositions 2.2 and 2.3 of [2], written with that paper's function $s_2 - s_3$, which is our g .

Proposition 3.1 (Prepared shifts and rarefaction). *For every sufficiently large N there are an integer L , numbers $\xi_j \in \{0, 1\}$ for $-J \leq j \leq J$, and a nonnegative integer $\zeta = \zeta_N$ with the following properties. Set*

$$d_j = \frac{3(3^{(j+1+J)\eta} - 1)}{2}, \quad A' = L + 2^\nu 3^\beta \mathbb{N}, \quad A'' = (L + 2^\nu 3^{\beta+\zeta} \mathbb{N}) \cap [N, 2N).$$

Then $L \equiv 9 \pmod{12}$, $d_j \equiv 0 \pmod{12}$, $A'' \subset A'$, and

$$(1) \quad g(n + d_j) - g(n) = jm + \xi_j \quad (n \in A', \quad -J \leq j \leq J).$$

If

$$I = \{k \in \mathbb{N} : N \leq L + 2^\nu 3^{\beta+\zeta} k < 2N\},$$

then, for every fixed $D > 0$, all but at most $C_D |I| \lambda^{-D}$ members $n \in A''$ satisfy

$$(2) \quad |g(n)| \leq Jm.$$

Moreover, $|I| \rightarrow \infty$ and $d_j = o(N)$ uniformly for $-J \leq j \leq J$.

The formula for d_j includes a terminal ternary zero: its base-3 expansion is a nonempty string of $(j+1+J)\eta$ ones followed by zero. Since η is divisible by four, $d_j \equiv 0 \pmod{12}$, including at $j = -J$.

4. A RESIDUE-UNIFORM FAIR-SHARE LEMMA

Proposition 2.4 of [2] treats the residue class $0 \pmod{m}$. Its proof gives every residue with the same error term.

Lemma 4.1 (Uniform fair share). *With A'' and I as in Proposition 3.1, for every residue $a \pmod{m}$,*

$$\#\{n \in A'' : g(n) \equiv a \pmod{m}\} = \frac{|I|}{m} (1 + o(1)),$$

where the $o(1)$ is uniform in a .

Proof. The parametrization of A'' by $k \in I$ is injective. Equation (52) of [2] expresses $g(n)$ on this progression as a constant shift of

$$H(k) = s_2(b_2 + 3^{\beta+\zeta}k) - s_3(2^\nu k),$$

with b_2 fixed. Write the constant so that $g(n) \equiv a \pmod{m}$ is equivalent to $H(k) \equiv t + a \pmod{m}$. Character orthogonality gives

$$(3) \quad P_a = \frac{|I|}{m} + \frac{1}{m} \sum_{1 \leq b < m} e\left(-\frac{b(t+a)}{m}\right) S_0(b/m),$$

where

$$S_0(\vartheta) = \sum_{k \in I} e\left(\vartheta s_2(b_2 + 3^{\beta+\zeta}k) - \vartheta s_3(2^\nu k)\right).$$

For $a = 0$, this is formula (61) of [2]. Replacing 0 by a only multiplies the summand at frequency b by $e(-ba/m)$, of modulus one.

The estimates (62)–(67) of [2] bound the absolute values of these same nonzero-frequency sums. They use, for $1 \leq b < m$, only $\|b/m\| \geq 1/m$ and absolute values of the exponential sums. The auxiliary parameters chosen after (67) therefore yield, independently of a ,

$$\sum_{1 \leq b < m} |S_0(b/m)| = o(|I|).$$

Taking absolute values in (3) gives

$$\left|P_a - \frac{|I|}{m}\right| \leq \frac{1}{m} \sum_{1 \leq b < m} |S_0(b/m)| = o\left(\frac{|I|}{m}\right),$$

uniformly in a . □

5. PROOF OF THE THEOREM

Proof of Theorem 2.1. Fix $z \in \mathbb{Z}$ and put $r = -z$. Choose any fixed $D > 1/2$ and let N tend to infinity through values for which Proposition 3.1 applies. By Lemma 4.1, with $a \equiv r \pmod{m}$, the number of $n \in A''$ in the required residue class is $|I|m^{-1}(1 + o(1))$. The number excluded by (2) is at most $C_D|I|\lambda^{-D}$. Relative to the fair-share main term this is

$$O_D(m\lambda^{-D}) = O_D\left(\frac{\lambda^{1/2-D}}{(\log \lambda)^{1/2+\epsilon}}\right) = o(1).$$

Thus, for all sufficiently large N , there is an $n \in A''$ such that $g(n) = r + jm$ and $|g(n)| \leq Jm$ for some $j \in \mathbb{Z}$. Since $m \rightarrow \infty$, assume also $m > |r|$. The last inequality then implies $-J \leq j \leq J$.

Because $A'' \subset A'$, (1) with index $-j$ applies. For $q = n + d_{-j}$ it gives

$$g(q) = g(n) - jm + \xi_{-j} = r + \xi_{-j} \in \{r, r+1\}.$$

Also $q \equiv L \equiv 9 \pmod{12}$. If $g(q) = r$, we are done. Otherwise $q \equiv 0 \pmod{3}$ and $q \equiv 1 \pmod{4}$. Adding one is carry-free in base 3, so $s_3(q+1) =$

$s_3(q) + 1$. In base 2, a number congruent to 1 (mod 4) ends in 01; adding one replaces this by 10, so $s_2(q + 1) = s_2(q)$. Hence $g(q + 1) = g(q) - 1 = r$.

Finally, $n \in [N, 2N)$ and $d_{-j} = o(N)$ uniformly in the allowed indices. The witness belongs to $[N, 2N + o(N) + 1]$. Taking successive sufficiently large N , each larger than the preceding witness, produces infinitely many distinct witnesses. Since $F = -g$, the theorem follows. $\square$

6. COMPUTATION, EXACT CHECKS, AND REPRODUCIBILITY

Computation played two different roles. A direct scan from 0 through 2,000,000 was used for discovery and for testing signs, examples, and likely offset behavior. This finite scan is not evidence for infinitude and is not used in the proof of Theorem 2.1.

Separate exact-integer checks audited the character indicator on finite cyclic groups, the implication bounding j , the formula and ternary representation of d_j , the congruences modulo 12, and the carry identity at $q \equiv 9 \pmod{12}$. These checks corroborate the displayed elementary identities but do not replace the analytic estimates imported from [2]; the theorem follows from the mathematical argument above, not from finite enumeration. The discovery program, an independent scan checker, the offset-reduction checker, and machine-readable exact outputs are supplied as ancillary material.

For methodological comparison, recent exact studies of other discrete sequences include a shifted-Golomb prefix identity proved through a conjugacy of run endpoints [4] and a generating-function proof of eventual monotonicity supplemented by exact finite checks [3]. Those works concern different sequences and are not inputs to the digit-sum argument above.

7. SCOPE AND LIMITATIONS

The theorem concerns nonnegative integers in bases 2 and 3, with $s_b(0) = 0$. It proves infinitude for each fixed integer offset, but the target is held fixed while $N \rightarrow \infty$. We do not claim uniformity as $|z|$ grows with N , an effective upper bound for the least witness, an asymptotic formula for a fixed fibre, or an extension to arbitrary pairs of multiplicatively independent bases.

The decisive analytic input remains Spiegelhofer's preparation and rarefaction constructions and exponential-sum estimates, together with the carry-propagation estimates used in his proof [2]. The argument here records the arbitrary-residue adaptation and its use in the endpoint argument. The ancillary finite computations neither certify those analytic estimates nor constitute a formal proof.

COMPUTATIONAL AND LANGUAGE-MODEL ASSISTANCE DISCLOSURE

Computer programs assisted exploratory enumeration and exact checking of elementary identities. Language-model systems assisted literature triage, proof development, adversarial checking, and drafting. A separate

language-model context independently reconstructed and challenged the decisive chain before manuscript preparation. Such assistance is not treated as mathematical authority: the proof presented here is self-contained apart from the explicitly cited inputs, and responsibility for the manuscript and its claims remains with the author.

REFERENCES

1. OEIS Foundation Inc., *Entry A180017: Difference of sums of digits of n in ternary and in binary*, The On-Line Encyclopedia of Integer Sequences, Accessed 4 September 2026.
2. Lukas Spiegelhofer, *Collisions of digit sums in bases 2 and 3*, Israel Journal of Mathematics **258** (2023), no. 1, 475–502, Author manuscript: arXiv:2105.11173v2 [math.NT], 20 April 2022.
3. Zijian Zeng, *The distinct-cycle-length probability is strictly decreasing after 30*, SSRN preprint, 2026.
4. ———, *A shifted-Golomb identity for the full-memory prefix of run-constructed almost Golomb sequences*, SSRN preprint, 2026.

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